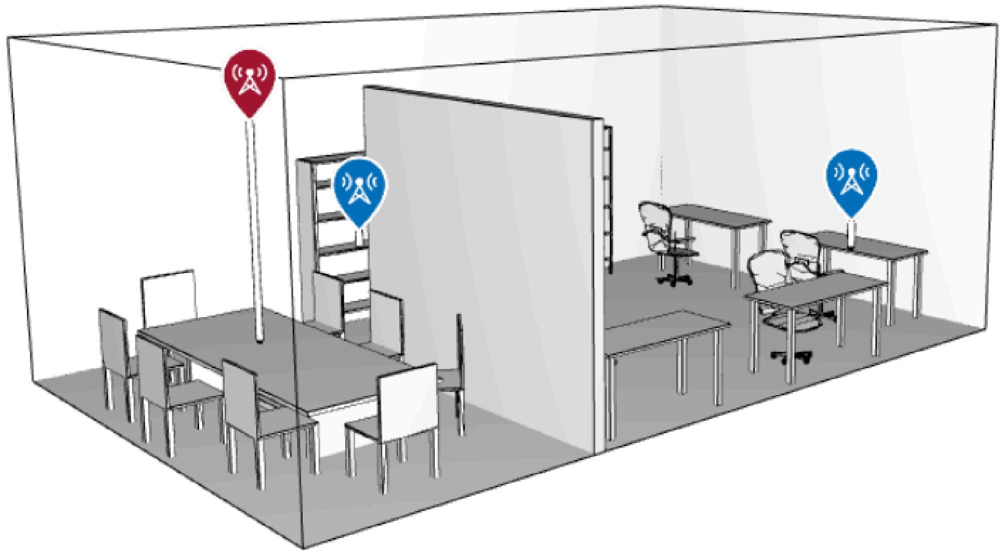
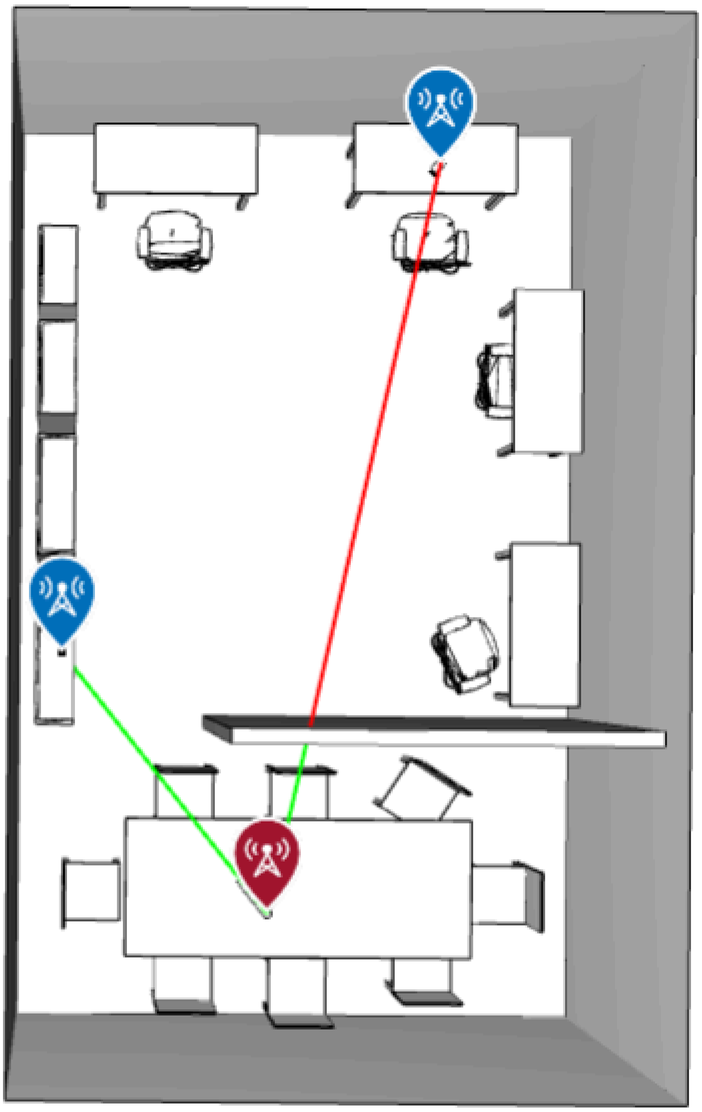
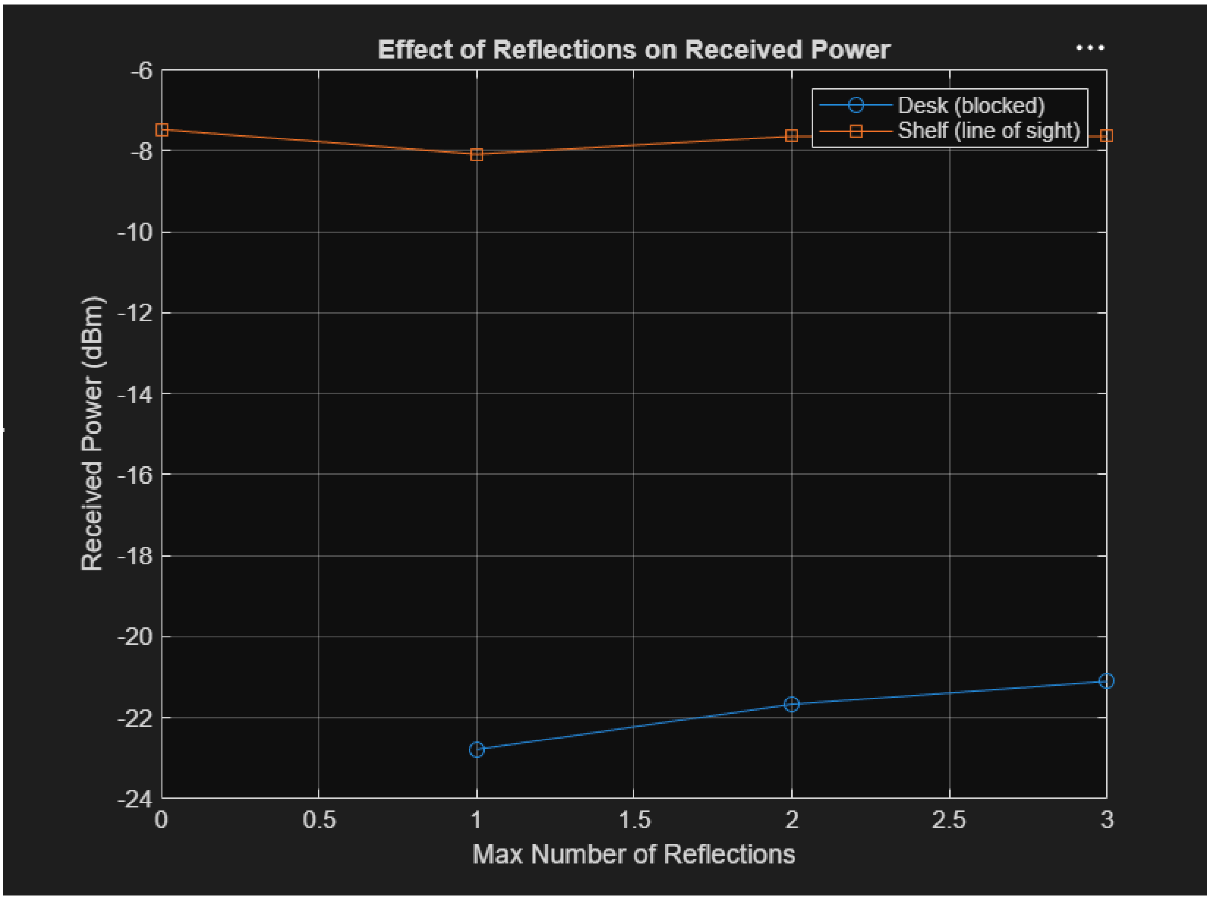
This project investigates how the number of reflections and diffractions affects predicted received power in an indoor MATLAB ray tracing scene, comparing a line-of-sight receiver against an obstructed one.

Setup:



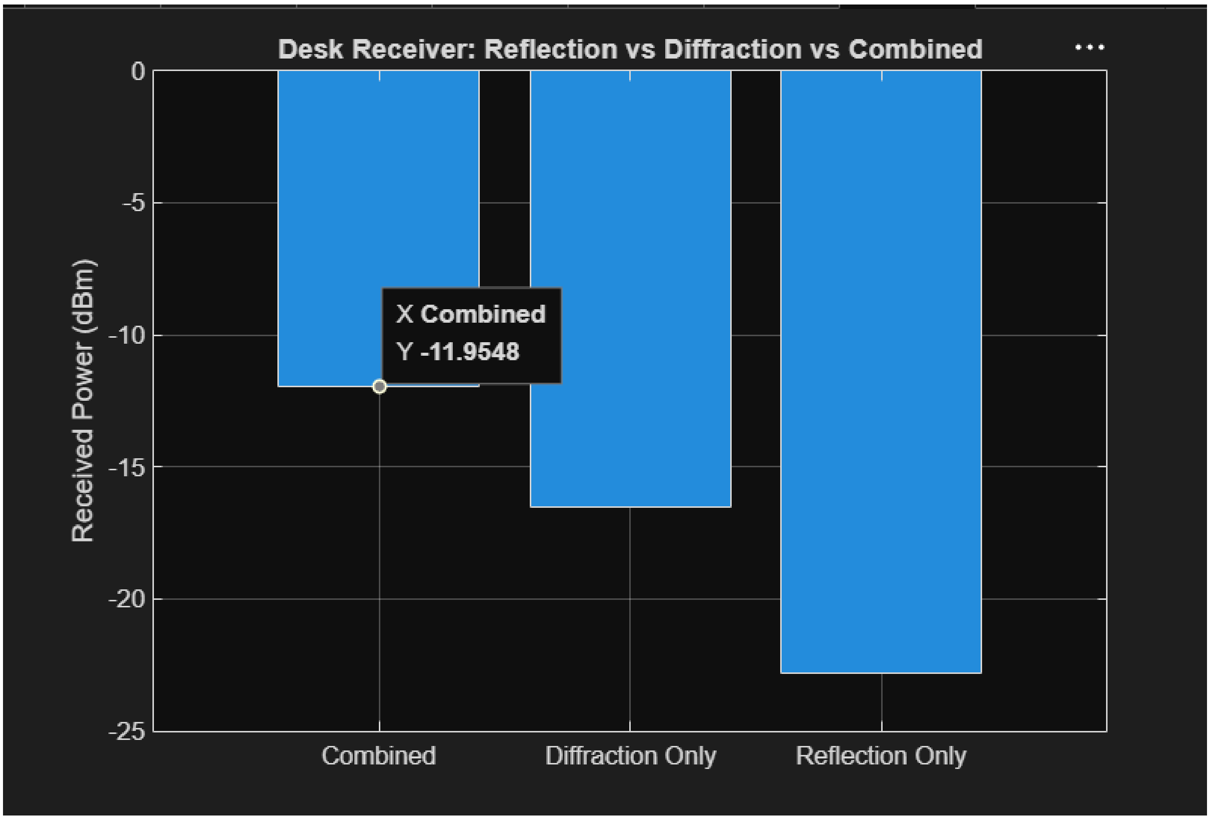


The green line to the shelf receiver indicates a clear, unobstructed path. The red line to the desk receiver indicates that the direct path is blocked by the partition wall this receiver will show -Inf received power until reflections or diffractions are enabled in the ray tracing model.



At 0 reflections, the desk receiver shows -Inf received power — no valid propagation path exists because the direct line is blocked by the partition wall. Enabling just 1 reflection immediately restores connectivity to -22.79 dBm. Increasing reflections further to 2 and 3 only improves the desk's received power marginally (to -21.67 and -21.11 dBm respectively), showing diminishing returns beyond the first reflection. The shelf receiver, which has clear line of sight, stays roughly stable around -7.5 to -8 dBm regardless of reflection count, since it doesn't depend on reflected paths to begin with.

```
Reflections=0 -> Desk=-Inf dBm | Shelf=-7.47 dBm
Reflections=1 -> Desk=-22.79 dBm | Shelf=-8.09 dBm
Reflections=2 -> Desk=-21.67 dBm | Shelf=-7.64 dBm
Reflections=3 -> Desk=-21.11 dBm | Shelf=-7.64 dBm
Reflection only   -> -22.79 dBm
Diffraction only  -> -16.53 dBm
Combined          -> -11.95 dBm
>>
```



To understand which propagation mechanism matters more for reaching the obstructed desk receiver, three cases were tested in isolation: reflection only (1 reflection, 0 diffractions), diffraction only (0 reflections, 1 diffraction), and both combined (1 reflection, 1 diffraction).

The results show that diffraction alone (-16.53 dBm) outperforms reflection alone (-22.79 dBm) by over 6 dB, indicating that the diffracted path around the partition's edge is a shorter or more direct route to the desk than the best available reflected path. When both mechanisms are enabled together, received power improves further to -11.95 dBm — stronger than either mechanism alone — showing that the reflected and diffracted paths contribute additively rather than one dominating the other.

This demonstrates that relying on reflection-only ray tracing settings, as is sometimes done to reduce computation time, can significantly underestimate coverage in scenes with sharp obstructions like partial walls. For accurate indoor coverage prediction near partitions or similar edges, enabling both reflections and diffractions is important, as their contributions combine rather than substitute for one another.